FINAL REPORT

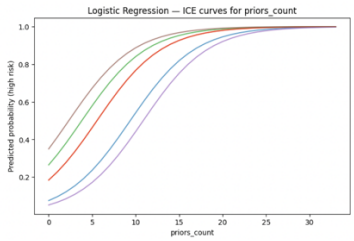
Part A – Data Stability and Drift Analysis

The analysis shows no meaningful distributional drift between the training and testing datasets across input features, encoded feature space, and model outputs. Key metrics confirm this stability: PSI values are well below 0.1, KS test p-values are high (indicating statistically indistinguishable distributions), and feature means remain consistent. In the transformed feature space, MMD^2 is effectively zero, confirming that preprocessing produces stable representations. At the prediction level, both Logistic Regression and Gradient-Boosted Tree (GBT) models exhibit nearly identical score distributions across train and test sets. Overall, these results confirm strong data alignment, absence of covariate shift, and reliable generalization conditions.

Part B – Generalization and Overfitting

Both models demonstrate good generalization with no severe overfitting, but differ in stability. Logistic Regression shows negligible performance gaps (accuracy gap ≈ 0.0017; AUC gap ≈ 0), indicating highly consistent and well-calibrated predictions. In contrast, GBT achieves slightly higher predictive performance but exhibits larger gaps (accuracy gap ≈ 0.025; AUC gap ≈ 0.0245), suggesting mild overfitting and higher sensitivity to training data. While still acceptable, this highlights a trade-off: GBT offers higher performance but lower stability, whereas Logistic Regression provides more robust and reliable generalization.

Part C – Spurious Correlation and Feature Sensitivity



Counterfactual analysis shows that both models are sensitive to categorical attributes, particularly race, but GBT exhibits larger probability shifts, indicating stronger dependence on these features. This suggests higher vulnerability to spurious correlations, especially for tree-based models.

Further analysis confirms that priors_count is the dominant predictor in both models. Predicted risk increases sharply as prior offenses rise, with probabilities exceeding 0.80 at high values. Logistic Regression shows a smooth, monotonic, and globally consistent relationship, while GBT exhibits non-linear, irregular, and heterogeneous responses, reflecting complex interactions but reduced stability.

Global Sensitivity Index results reinforce this finding: both models are highly sensitive to priors_count, though Logistic Regression shows a slightly stronger and more uniform effect. Overall, this highlights a key trade-off:

- Logistic Regression → interpretability + stability
- GBT → flexibility + interaction modeling (but higher risk of instability)

Part D – Fairness and Subgroup Analysis

Slice-based evaluation reveals significant disparities across demographic groups, particularly by age and race. Both models perform best for individuals aged 25–45, while younger individuals (<25) have higher false positives and older individuals (>45) have higher false negatives, indicating calibration issues.

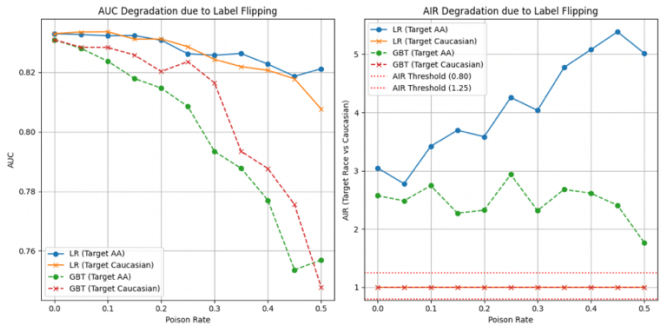
Gender disparities are moderate, but racial disparities are most pronounced: African-American individuals show higher predicted risk and false positive rates, while Caucasian and Hispanic groups have more balanced performance. Smaller subgroups show unstable results due to limited sample size.

GBT provides slightly higher performance in some segments but shows greater variability, while Logistic Regression remains more consistent across groups. These findings highlight critical fairness concerns requiring subgroup-level monitoring and mitigation.

Part 1 – PGD Evasion Audit

Both models behave similarly under adversarial perturbations to priors_count, showing minimal changes in FPR and AIR. However, some groups (e.g., Asian and Other) consistently exhibit AIR < 0.80, indicating pre-existing fairness disparities. The attack does not significantly worsen or improve these biases, suggesting that fairness issues stem from the data rather than adversarial manipulation.

Part 2 – Label-Flipping (Poisoning) Attack



Label-flipping causes gradual AUC degradation, with GBT deteriorating faster, indicating higher vulnerability. A key finding is the existence of a "stealth zone" (≈5–10% poisoning) where:

- AUC drops minimally (<2%)
- Fairness (AIR) is significantly distorted beyond acceptable bounds

This makes attacks difficult to detect using performance metrics alone. Additionally, since only labels are altered, traditional drift metrics (e.g., PSI) fail to detect the issue. This demonstrates that fairness can be manipulated without noticeable performance loss, posing a critical real-world risk.

Part 3 – Privacy (Membership Inference Attacks)

Both models show strong privacy robustness, with MIAUC ≈ 0.5 (random guessing). Logistic Regression has near-zero generalization gap, while GBT shows mild overfitting, but neither leads to increased attack vulnerability. Confidence gap analysis confirms that Logistic Regression produces more stable predictions, while GBT is more variable.

L2 regularization has minimal impact on privacy risk, as MIAUC remains unchanged across different C values. This indicates that regularization is not a critical factor for mitigating membership inference attacks in this setting.

Highest-Risk Finding & Mitigation

Highest Risk: Stealth Zone in Label-Flipping Attacks

The most critical risk is the stealth zone, where fairness is severely compromised without triggering performance alerts. This allows attackers to silently manipulate model fairness, especially for sensitive groups.

Mitigation Strategies

1. Proactive – L2 Regularization

Stronger regularization slightly improves robustness and stabilizes AUC but does not eliminate fairness distortion. Its overall impact is limited, though it introduces minimal disparate impact.

2. Reactive – Disaggregated Fairness Monitoring

Monitoring fairness metrics (AIR, FPR, FNR) by subgroup is highly effective. It detects early fairness violations even when AUC remains stable, making it the most practical defense.

Trade-offs

- Regularization improves robustness with minimal downside
- Fairness monitoring may flag certain groups more often, but this is beneficial for identifying bias
- Addressing root causes of bias is more important than model selection alone

Overall, both models are stable, generalizable, and privacy-robust, but fairness remains the primary risk, particularly under adversarial conditions. The Gradient-Boosted Tree offers higher performance but is more sensitive to noise, bias, and attacks, while Logistic Regression provides greater stability and interpretability. The presence of a stealth zone highlights a critical limitation of traditional evaluation methods, emphasizing the need for integrated monitoring frameworks that jointly assess performance, fairness, and robustness.